

Computer Networks (CS-307) – Assignment #01

Instructor: Mr. Muhammad Shakeel	BSAI-6B / BSIT-6
Submission Date: 02-03-2026 09:30 AM	Marks: 20

Submission Guidelines:

- Build all topologies in Cisco Packet Tracer and save .pkt files as: Your Name_Roll_ TaskNo .pkt
- Take clear, labeled screenshots of topology view and simulation results for each task.
- Write a short lab report section for each task explaining your observations.
- Your Answer must be properly commented.
- Try NOT to copy-paste data from your friends, if you want to improve your **Networking Skills do it by yourself**.
- This is an individual assignment. **PLAGIARISM IS NOT ACCEPTABLE!** In case of plagiarism, you will get **ZERO MARKS** for that question
- Marks will be deducted if the Answer does not conform to the standards set out in the document **Computer Network for Assessed Work**.
- **Submit files on your LMS Account.**
- **Online Submission date 01-03-2026 11:59 PM.**

Read all instructions carefully before starting.

TASK 01: Hub-Based Network — Bus Topology

Topology: Bus Topology | Device: Hub (Generic)

Background Concept

A Hub is a Layer 1 (Physical Layer) device that has no intelligence — it simply broadcasts every incoming frame to ALL connected ports regardless of the destination. When combined with Bus Topology (all devices on a single backbone cable), the entire network becomes one large collision domain. Any two devices transmitting at the same time cause a collision, forcing CSMA/CD to resolve it. Network performance degrades rapidly as more devices are added.

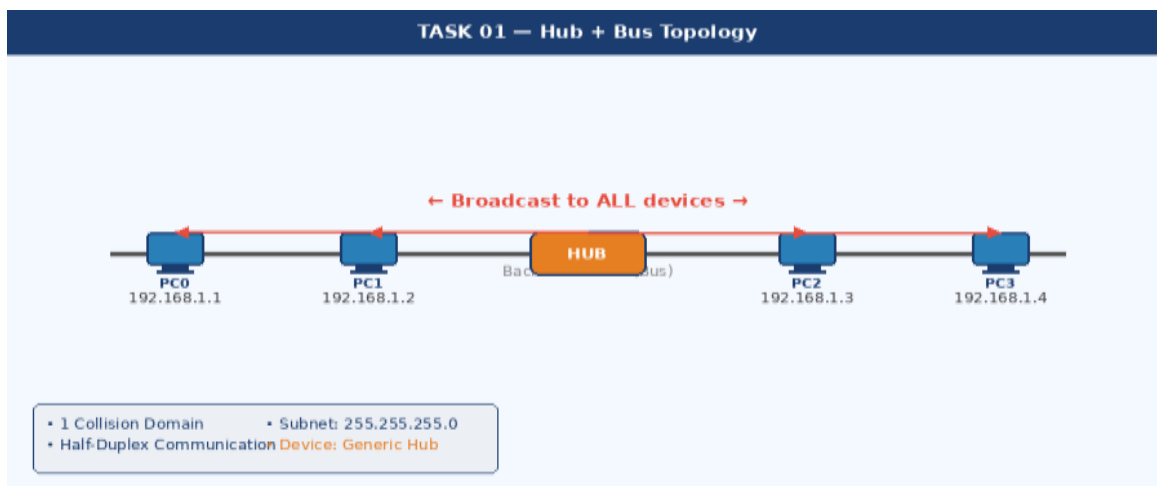


Figure 1: Hub + Bus Topology — Packet broadcasts to all devices

Network Setup

- Devices: 1 Hub + 4 PCs | Cables: Copper Straight-Through
- PC0: 192.168.1.1 | PC1: 192.168.1.2 | PC2: 192.168.1.3 | PC3: 192.168.1.4
- Subnet Mask for all PCs: 255.255.255.0 | Layout: Horizontal bus-style line

Step-by-Step Procedure

1. Open Packet Tracer. From Network Devices > Hubs, drag a Generic Hub onto the workspace.
2. Add 4 PCs from End Devices. Arrange them in a horizontal line.
3. Connect each PC to the Hub using Copper Straight-Through cable.
4. Click each PC > Desktop > IP Configuration. Enter the IP address and subnet mask as listed above.
5. Switch to Simulation Mode. Set PDU filter to ICMP only.
6. From PC0 > Desktop > Command Prompt, type: ping 192.168.1.3 and press Enter.
7. Click Play. Observe the packet broadcast — it reaches ALL 4 PCs, not just PC2.
8. Take screenshots of the completed topology and the simulation broadcast event.

Observations to Write in Report

- How many devices received the packet even though you pinged only one? Why?
- How many collision domains exist in this network? Explain.
- Explain CSMA/CD in your own words based on what you observed.

Deliverables

- Screenshot: Completed Bus Topology with Hub
- Screenshot: Simulation showing broadcast to all PCs
- Saved .pkt file | Lab report: min. 150 words

TASK 02: Hub-Based Network — Star Topology

Topology: Star Topology | Device: Hub (Generic)

Background Concept

In Star Topology every device has its own dedicated cable connecting to a central device. Even though the physical layout looks like a star, a Hub as the central device means the logical behavior is identical to Bus — it still broadcasts to ALL ports. This is an important distinction: physical topology (cable layout) vs logical topology (data flow) can be completely different. A Hub-based Star is still one single collision domain with no improvement over Bus Topology in terms of collision handling.

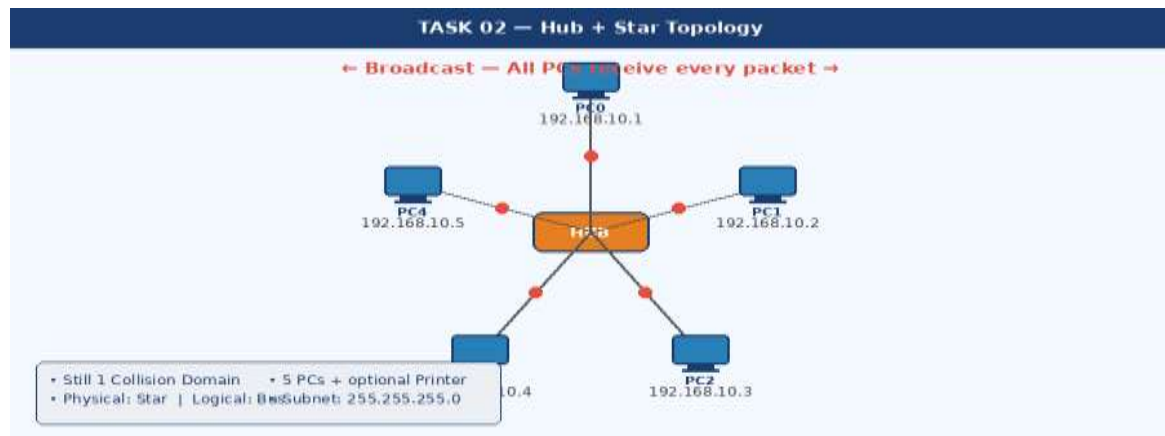


Figure 2: Hub + Star Topology — Hub still broadcasts to every connected PC

Network Setup

- Devices: 1 Hub + 5 PCs | Cables: Copper Straight-Through
- PC0: 192.168.10.1 | PC1: 192.168.10.2 | PC2: 192.168.10.3 | PC3: 192.168.10.4 | PC4: 192.168.10.5
- Subnet Mask: 255.255.255.0 | Layout: Hub at center, PCs around it in star pattern
- BONUS (+5 marks): Add a Printer with IP 192.168.10.6 also connected to the Hub

Step-by-Step Procedure

9. Place 1 Hub at the center. Add 5 PCs arranged around it in a star/circular pattern.
10. Connect all 5 PCs to Hub using Straight-Through cables.
11. Assign IP addresses to all PCs as listed above.
12. Simulation Mode: Add a Simple PDU from PC0 to PC4. Press Play.
13. Observe: packet broadcasts to ALL PCs — PC1, PC2, PC3 also receive it.
14. Try two simultaneous PDUs (PC0→PC3 and PC1→PC4) and observe collision behavior.

Observations & Comparison

- Was there any difference in packet delivery between Task 1 (Bus) and Task 2 (Star)?
- How many collision domains exist in a Hub-based Star network with 5 PCs?
- What is the difference between physical topology and logical topology?

Deliverables

- Screenshot: Star topology with Hub and 5 PCs | Screenshot: Simulation broadcast
- Saved .pkt file | Lab report comparing Hub in Bus vs Star: min. 200 words

TASK 03: Switch-Based Network — Bus Topology

Topology: Bus Topology | Device: Cisco 2960 Switch

Background Concept

A Switch is a Layer 2 (Data Link Layer) device that uses a MAC Address Table to intelligently forward frames only to the port where the destination device is connected. Each Switch port is its own collision domain, enabling full-duplex communication. The Switch builds its MAC table dynamically — when a device first sends data, the Switch records its MAC address and port number. This intelligent forwarding eliminates unnecessary traffic and dramatically improves network performance.

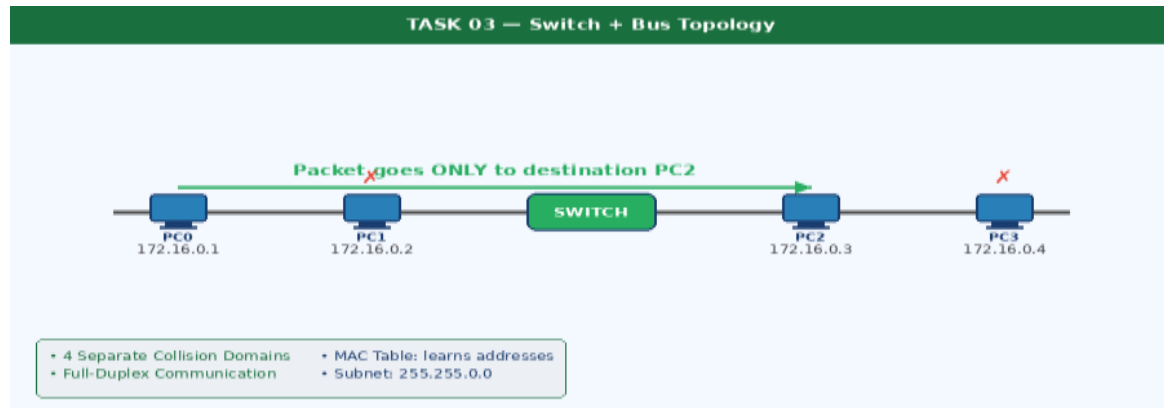


Figure 3: Switch + Bus Topology — Packet delivered ONLY to destination

Network Setup

- Devices: 1 Cisco 2960 Switch + 4 PCs | Cables: Copper Straight-Through
- PC0: 172.16.0.1 | PC1: 172.16.0.2 | PC2: 172.16.0.3 | PC3: 172.16.0.4
- Subnet Mask: 255.255.0.0 | Layout: Horizontal bus-style arrangement

Step-by-Step Procedure

15. Add a Cisco 2960 Switch from Network Devices > Switches.
16. Add 4 PCs in horizontal line. Connect all to Switch via Straight-Through cable.
17. Assign IP addresses to all 4 PCs as listed above.
18. Simulation Mode. From PC0 > Command Prompt: ping 172.16.0.3
19. Observe: packet goes ONLY to PC2. PC1 and PC3 do NOT receive it. Screenshot this.
20. Click Switch > CLI > type: enable then: show mac-address-table. Screenshot the output.

Observations & Analysis

- What was the key difference between Hub (Task 1) and Switch (Task 3) packet delivery?
- How does the Switch learn and store MAC addresses? Explain the learning process.
- What is ARP? Why was the first ping slower than later pings?
- How many collision domains exist in a Switch network with 4 PCs?

Deliverables

- Screenshot: Bus-layout topology with Switch
- Screenshot: Simulation showing selective delivery to destination only
- Screenshot: MAC Address Table from Switch CLI (show mac-address-table)
- Saved .pkt file | Lab report: Hub vs Switch comparison — min. 200 words

TASK 04: Switch — Star Topology + VLAN Configuration

Topology: Star Topology | Device: Cisco 2960 Switch (Advanced)

Background Concept

VLANs (Virtual LANs) allow you to logically segment one physical network into multiple isolated groups. Devices in different VLANs cannot communicate even if connected to the same Switch. This is a core enterprise networking feature used to separate departments (Admin, Finance, HR) and improve security. In this task you will create 2 VLANs on a Cisco 2960 Switch, assign ports to each VLAN, and verify that communication works within a VLAN but is blocked between VLANs.

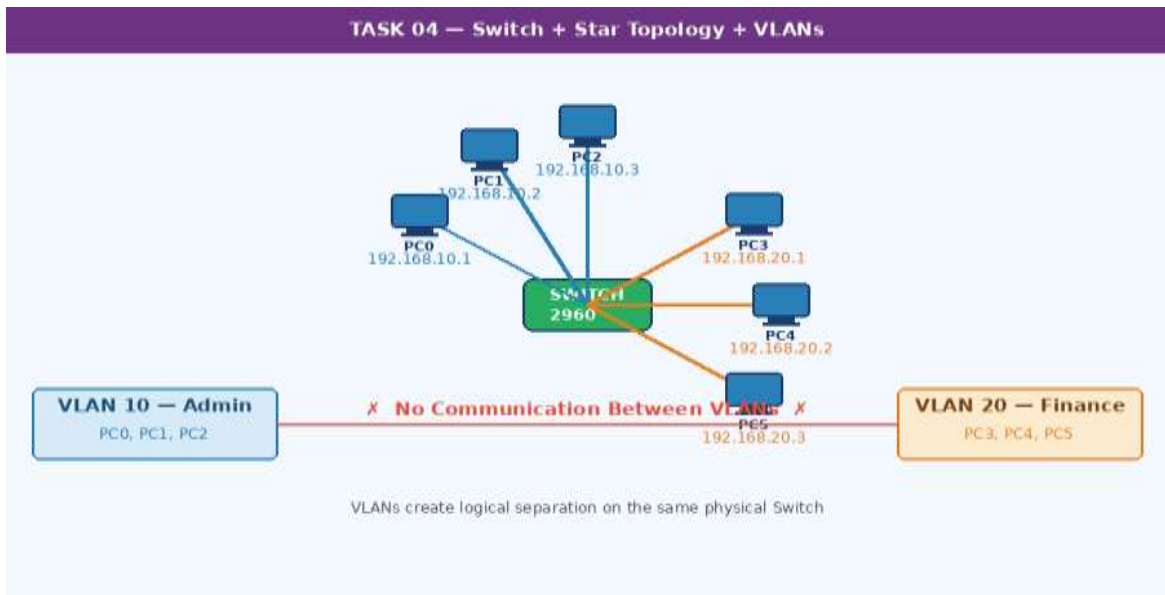


Figure 4: Switch + Star Topology with VLAN 10 (Admin) and VLAN 20 (Finance) isolated

Network Setup

- Devices: 1 Cisco 2960 Switch + 6 PCs | Layout: Star — Switch at center
- VLAN 10 (Admin): PC0-Fa0/1 | PC1-Fa0/2 | PC2-Fa0/3
- VLAN 20 (Finance): PC3-Fa0/4 | PC4-Fa0/5 | PC5-Fa0/6
- PC0: 192.168.10.1 PC1: 192.168.10.2 PC2: 192.168.10.3 (VLAN 10 — Subnet: 255.255.255.0)
- PC3: 192.168.20.1 PC4: 192.168.20.2 PC5: 192.168.20.3 (VLAN 20 — Subnet: 255.255.255.0)

Testing Steps

21. Verify VLANs with: show vlan brief — Screenshot this output.
22. From PC0, ping 192.168.10.2 (PC1) — This SHOULD succeed (same VLAN).
23. From PC0, ping 192.168.20.1 (PC3) — This SHOULD FAIL (different VLAN).
24. Screenshot both ping results clearly.

Deliverables

- Screenshot: Star topology with 6 PCs and Switch | Screenshot: show vlan brief output
- Screenshot: Successful ping within VLAN 10 | Screenshot: Failed ping between VLANs
- Saved .pkt file | Lab report: VLAN concept and observations — min. 300 words